

Fractional mass does not control flat filling

A scoped counterexample for the weighted polygonal closure in arXiv:2403.03901

Automated discovery run `fa_banach_001`, lane 6

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Status. Candidate counterexample, likely valid, pending expert review. The result answers the source question negatively for the closure generated by the source paper’s *weighted* polygonal approximants. A stricter closure using only integer coefficients remains outside the claim.

1 Source question

The source is Cicalese–Heilmann–Kubin–Onoue–Ponsiglione, *A notion of s -fractional mass for 1-currents in higher codimension*, arXiv:2403.03901v1 [1]. In the Conclusions (printed page 24), after suggesting that “integer” currents might be defined as a closure of polyhedral currents in the convergence of Theorem 5.1, the authors ask whether a closed current of finite fractional mass bounds a 2-current of bounded local mass, and whether the fractional mass controls the flat norm.

line tension models, as $\varepsilon \rightarrow 0$ [11, 8]. We also remark that, for technical reasons, it could be convenient to replace the denominator $\max\{|x - y|, \varepsilon\}$ with a function which is strictly increasing also around zero. The study of the limit as $s \rightarrow 0$ of the s -fractional mass could also deserve some attention.

As explained in the Introduction, the proposed notion of s -fractional mass generalizes that of s -fractional perimeters of sets in $\mathbb{R}^2$ to higher codimension and not necessarily integer rectifiable currents. In fact, as for sets with finite fractional perimeter, there is no reason to expect that the 1-currents with bounded fractional mass are rectifiable; now, a natural question arises: what are integer (not necessarily rectifiable) currents in this context? One possibility is to exploit our Smirnov type approximation result, defining the integer currents with finite mass as the closure of polyhedral 1-currents with respect to the notion of convergence in Theorem 5.1. Now, take an integer (i.e., limit of smooth curves) 1-current γ without boundary and with finite s -fractional mass; is there a 2-current, with bounded area (i.e., bounded local mass) whose boundary is γ ? Furthermore, is the mass controlled by the s -fractional mass of its boundary? In the terminology of geometric measure theory, we ask if the fractional mass controls the flat norm. In the flat case of 1-currents in $\mathbb{R}^2$ this is true, being nothing but the fractional isoperimetric inequality. A positive answer to the questions posed above would suggest that integer 1-currents with bounded fractional mass are a good class of boundary data for setting up the Plateau problems for minimal surfaces.

Let us also mention that our definition of higher codimensional fractional mass could be generalized to the case of k -forms in $\mathbb{R}^d$ for all $1 \leq k \leq d$. Finally, also the case $k = 0$ corresponding to charged masses could be investigated.

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Figure 1: The source question on printed page 24 of arXiv:2403.03901v1.

The distinction between weighted and integer coefficients matters. Theorem 5.1 and Corollary 5.3 of the source approximate closed elements of $\mathcal{C}_s$ by closed polygonal currents with coefficients $\delta_n \rightarrow 0$, not by integer-coefficient currents. The result below treats exactly the closure available from that approximation theorem.

2 Definitions and statement

Fix $d \geq 3$ and $0 < s < 1$, and put

$$\beta = d - s, \quad \theta = \frac{\beta - 2}{2} = \frac{d - s - 2}{2} > 0.$$

For a compactly supported 1-current γ whose Fourier transform is a function, the source norm is, up to a positive constant depending only on d, s ,

$$\|\gamma\|_{\mathcal{C}_s}^2 \asymp \int_{\mathbb{R}^d} |\widehat{\gamma}(\xi)|^2 |\xi|^{-\beta} d\xi, \quad M_s(\gamma) = (2\pi)^{-d} \|\gamma\|_{\mathcal{C}_s}^2.$$

A 2-current S has locally finite mass if its mass measure is finite on every compact set. If $\partial S = \gamma$, then for every compactly supported smooth 1-form φ ,

$$|\gamma(\varphi)| = |S(d\varphi)| \leq \mathbf{M}(S \llcorner K) \|d\varphi\|_{\infty}, \quad (1)$$

whenever K contains $\text{supp}(d\varphi)$.

Theorem 1 (No flat filling in the weighted closure). *For every $d \geq 3$ and $0 < s < 1$ there are a compact set $K \subset \mathbb{R}^d$ and a closed 1-current $\gamma \in \mathcal{C}_s$, supported in K , with the following properties.*

1. *There are weighted closed polygonal 1-currents P_j such that $P_j \rightarrow \gamma$ weakly as currents and $M_s(P_j) \rightarrow M_s(\gamma)$.*
2. *There is no 2-current of locally finite mass S with $\partial S = \gamma$.*

Consequently, the closure generated by the weighted approximation in Theorem 5.1 and Corollary 5.3 of the source does not have the proposed filling property, and M_s cannot control its flat norm.

There is also a quantitative obstruction before taking the infinite sum.

Proposition 1. *There is a sequence of smooth compactly supported divergence-free vector fields γ_N , all supported in one compact set, such that*

$$\sup_N M_s(\gamma_N) < \infty, \quad \inf_{\partial S = \gamma_N} \mathbf{M}(S \llcorner K) \rightarrow \infty,$$

where the infimum is over 2-currents of locally finite mass.

3 Proof intuition

The norm M_s is a negative-order Fourier norm in dimensions $d \geq 3$. Writing a cycle as the boundary of a smooth 2-current with density $A \sin(Nx_1)$ costs one derivative, but the source weight returns $d - s$ derivatives in the squared norm. The resulting $\mathcal{C}_s$ norm is only $O(AN^{-\theta})$, whereas a compactly supported flat-dual test sees the full potential amplitude A . Thus a high frequency hides arbitrarily large filling cost from fractional mass.

For the single counterexample, choose amplitudes tending to infinity and frequencies so lacunary that the corresponding $\mathcal{C}_s$ norms and all unwanted cross pairings are summable. The cycle series converges in $\mathcal{C}_s$, while a sequence of uniformly admissible flat-dual tests has pairings tending to infinity. Local finite mass of a filling would uniformly bound all those pairings on their common compact support, which is impossible.

4 Proof

Choose nonzero real functions $a \in C_c^\infty(\mathbb{R})$ and $b \in C_c^\infty(\mathbb{R}^{d-1})$. Write $x' = (x_2, \dots, x_d)$ and $e_{12} = e_1 \wedge e_2$. For $A > 0$ and an integer $N \geq 2$, define the smooth 2-current

$$T_{A,N} = u_{A,N}(x) e_{12} dx, \quad u_{A,N}(x) = A, a(x_1) b(x') \sin(Nx_1),$$

and let $\gamma_{A,N} = \partial T_{A,N}$. In vector notation,

$$\gamma_{A,N} = (\partial_2 u_{A,N}, -\partial_1 u_{A,N}, 0, \dots, 0),$$

so it is smooth, compactly supported, and divergence-free.

Lemma 1 (Shift estimate). *There is a constant C , depending only on a, b, d, s , such that*

$$\|\gamma_{A,N}\|_{\mathcal{C}_s} \leq C A N^{-\theta} \quad \text{for all } A > 0, N \geq 2.$$

Proof. Let $\eta(x) = a(x_1) b(x')$. Fourier transformation gives

$$|\widehat{\gamma}_{A,N}(\xi)|^2 = (\xi_1^2 + \xi_2^2) |\widehat{u}_{A,N}(\xi)|^2, \quad \widehat{u}_{A,N}(\xi) = \frac{A}{2i} (\widehat{\eta}(\xi - Ne_1) - \widehat{\eta}(\xi + Ne_1)).$$

It is enough to estimate either translate. After writing $\zeta = \xi \mp Ne_1$, split the integral into $|\zeta| \leq N/2$ and its complement. On the first region,

$$\frac{\xi_1^2 + \xi_2^2}{|\xi|^\beta} \leq |\xi|^{2-\beta} \leq C N^{2-\beta}.$$

On the complementary region, the Schwartz decay of $\widehat{\eta}$ gives the same bound (indeed an error smaller than every power of N). The possible singularity at $\xi = 0$ is locally integrable because $|\xi|^{2-\beta} = |\xi|^{2-d+s}$. Hence

$$\|\gamma_{A,N}\|_{\mathcal{C}_s}^2 \leq C A^2 N^{2-\beta} = C A^2 N^{-2\theta},$$

which proves the claim. $\square$

Lemma 2 (Uniform flat-dual tests). *There are compactly supported smooth 1-forms φ_N , all supported in one compact set, and constants $c, C > 0$ such that*

$$\|d\varphi_N\|_\infty \leq 1, \quad |\gamma_{A,N}(\varphi_N)| \geq cA$$

for all sufficiently large N . Moreover, for every $L > 0$ and $m \neq n$,

$$\begin{aligned} |\gamma_{A_m, N_m}(\varphi_{N_n})| \\ \leq C_L A_m ((1 + |N_m - N_n|)^{-L} + (1 + N_m + N_n)^{-L}). \end{aligned} \quad (2)$$

Proof. For a sufficiently small constant $c_0 > 0$, set

$$\varphi_N = c_0 N^{-1} a(x_1) b(x') \cos(Nx_1) dx_2.$$

All first derivatives of its coefficient are uniformly bounded, and choosing c_0 small gives $\|d\varphi_N\|_\infty \leq 1$. Since $T_{A,N}$ has only an e_{12} component,

$$\begin{aligned} \gamma_{A,N}(\varphi_N) &= T_{A,N}(d\varphi_N) \\ &= c_0 A \int a(x_1) b(x') \sin(Nx_1) (N^{-1} a'(x_1) b(x') \cos(Nx_1) - a(x_1) b(x') \sin(Nx_1)) dx. \end{aligned}$$

The second term tends to $-\frac{c_0 A}{2} \|a\|_2^2 \|b\|_2^2$, while the first is $o(A)$. This proves the diagonal lower bound. For different frequencies, the same formula is a finite sum of Fourier coefficients of the smooth compactly supported functions a^2 and aa' , at frequencies $N_m - N_n$ and $N_m + N_n$. Their rapid decay proves (2). $\square$

Proposition 1 follows immediately: take $A_N = N^\theta$. Lemma 1 gives uniformly bounded fractional mass, while Lemma 2 and (1) force every filling to have mass at least cN^θ on the common compact support.

We now build one nonfillable current. Put $A_j = j$. Using Lemmas 1 and 2, choose integers N_j inductively, as lacunary as necessary, so that

$$\|\gamma_{A_j, N_j}\|_{\mathcal{C}_s} \leq 2^{-j} \quad (3)$$

and, for every $i < j$,

$$|\gamma_{A_j, N_j}(\varphi_{N_i})| + |\gamma_{A_i, N_i}(\varphi_{N_j})| \leq c_1 2^{-i-j-4}, \quad (4)$$

where $c_1 > 0$ is smaller than the diagonal constant in Lemma 2. Such a choice is possible because $\theta > 0$ and every off-diagonal pairing tends to zero faster than any power as the new frequency tends to infinity.

By (3), the series

$$\gamma = \sum_{j=1}^{\infty} \gamma_{A_j, N_j}$$

converges in the weighted Fourier norm. It also converges as a current by weighted Cauchy–Schwarz against the Fourier transform of each test form. All summands have support in one compact set, so the limit is compactly supported; its Fourier transform is the weighted L^2 limit, hence $\gamma \in \mathcal{C}_s$. Also $\partial\gamma = 0$.

The diagonal estimate and (4) give

$$|\gamma(\varphi_{N_j})| \geq c_1 A_j - \sum_{m \neq j} |\gamma_{A_m, N_m}(\varphi_{N_j})| \geq \frac{c_1}{2} j$$

after decreasing c_1 if necessary. Thus these pairings are unbounded, although $\|d\varphi_{N_j}\|_\infty \leq 1$ and all $d\varphi_{N_j}$ have common compact support. Equation (1) rules out every locally finite-mass 2-current S with $\partial S = \gamma$.

Finally, Corollary 5.3 of the source applies to every closed $\gamma \in \mathcal{C}_s$ and supplies weighted closed polygonal currents converging weakly to γ with convergence of M_s . This proves both assertions of Theorem 1. $\square$

5 Verification notes and limitations

The proof is symbolic. Its two threshold checks are sharp for this method: $\theta > 0$ exactly when $d - s > 2$, and the source explicitly records the planar case as positive. The singular part of the Fourier estimate is integrable, and translated Schwartz decay controls the only region where the simple $N^{2-\beta}$ pointwise estimate fails.

The conclusion is stronger than failure of a uniform bound: the lacunary limit has no filling of locally finite mass at all. On the other hand, the coefficient caveat is essential. The proof uses the weighted closure provided by Corollary 5.3. It does not prove that γ is a limit of integer-coefficient polyhedral currents in the same topology. Therefore a classically integral-coefficient version of the source question remains open.

Human review should focus on Lemma 1, the simultaneous lacunary choice in (3)–(4), and the intended meaning of “integer” in the source Conclusions.

6 Novelty search

On August 27, 2026, a bounded search covered the run’s registry, solution, attempt, and proof-gap indexes; arXiv:2403.03901; the exact title and author combination; and arXiv searches combining **fractional mass**, **flat norm**, **weighted polygonal**, **Plateau**, and **1-currents**. The arXiv record has only v1. No later primary-source answer or the oscillatory/lacunary obstruction above was found. This is not a claim of exhaustive priority.

References

- [1] M. Cicalese, T. Heilmann, A. Kubin, F. Onoue, and M. Ponsiglione, *A notion of s -fractional mass for 1-currents in higher codimension*, arXiv:2403.03901v1, 2024.